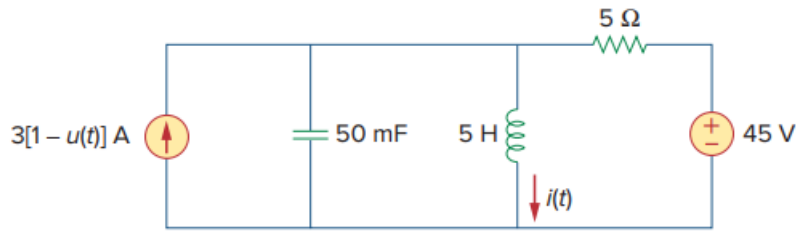


Problem

Determine $i(t)$ for $t > 0$ in the circuit of Fig. 8.96.

Figure 8.96:



Step-by-step solution

Step 1 of 13

Refer to Figure 8.96 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 3 A since the value $u(t)$ at $t < 0$ is 0 .

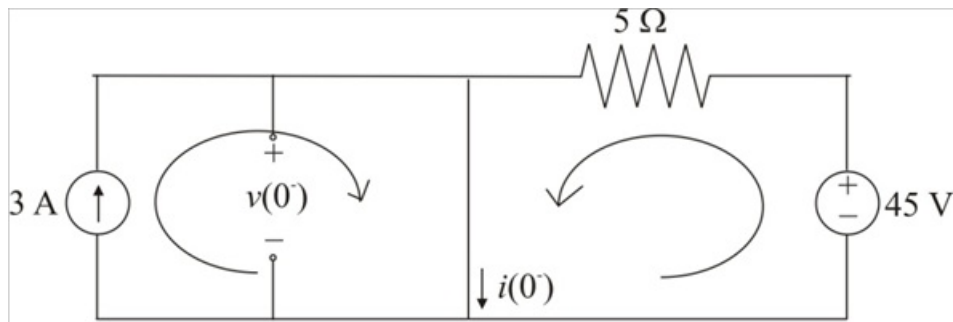


Figure 1

Step 2 of 13

The value of the voltage $v(0^-)$ is 0 V since both the terminals are shorted.

Calculate the value of the current $i(0^-)$.

$$\begin{aligned} i(0^-) &= 3 + \frac{45}{5} \\ &= 3 + 9 \\ &= 12 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 12 A .

Step 3 of 13

Draw the circuit diagram at $t > 0$.

The value of the current source is 0 A since $u(t)$ is 1 at $t > 0$.

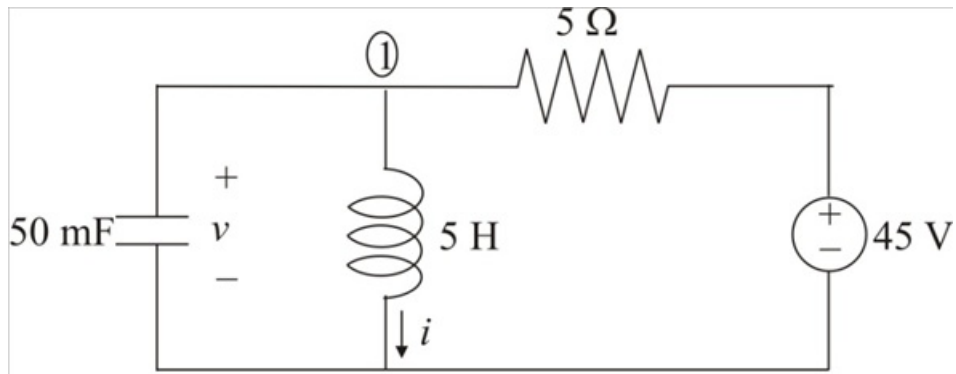


Figure 2

Step 4 of 13

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 12 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 0\text{ V}$ and $i(0) = 12\text{ A}$.

Step 5 of 13

Calculate the value of the voltage v in terms of the inductance.

$$v = 5 \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{v}{5}$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = \frac{v(0)}{5}$$

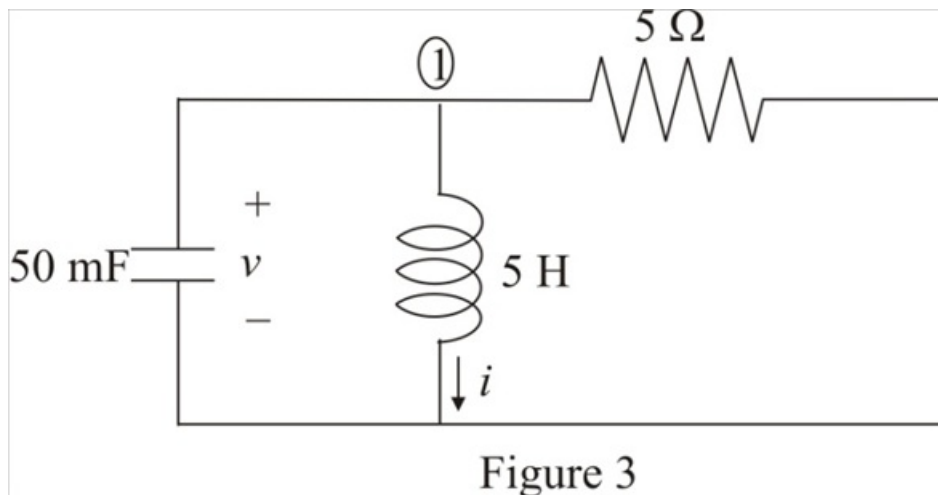
Substitute 0 V for $v(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= \frac{0}{5} \\ &= 0\text{ A/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is 0 A/s .

Step 6 of 13

Deactivate the sources from the circuit diagram to estimate the transient response.



Step 7 of 13

Apply Kirchhoff's Current Law at node-1.

$$\frac{v}{5} + i + (50 \text{ m}) \frac{dv}{dt} = 0$$

Substitute $5 \frac{di}{dt}$ for v .

$$\frac{1}{5} \left(5 \frac{di}{dt} \right) + i + (50 \text{ m}) \frac{d}{dt} \left(5 \frac{di}{dt} \right) = 0$$

$$\frac{di}{dt} + i + 0.25 \frac{d^2 i}{dt^2} = 0$$

$$\frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 4i = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2 i + 4s i + 4i = 0$$

$$(s^2 + 4s + 4)i = 0$$

Step 8 of 13

Consider the characteristic equation of the circuit.

$$s^2 + 4s + 4 = 0$$

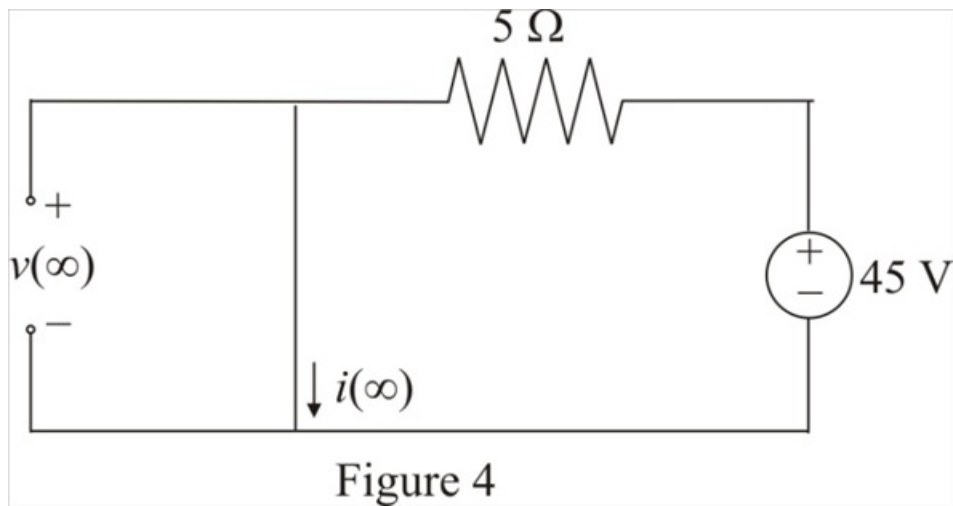
$$(s + 2)(s + 2) = 0$$

Therefore, the roots of the equation are -2 and -2 .

The circuit exhibits critical damped response since the roots of the characteristic equation are negative and equal.

Step 9 of 13

Draw the circuit for $t = \infty$.



Step 10 of 13

Calculate the value of the current $i(\infty)$.

$$\begin{aligned} i(\infty) &= \frac{45}{5} \\ &= 9 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(\infty)$ is 9 A.

Step 11 of 13

Write the general solution for the voltage $v(t)$ for under-damped condition.

$$i(t) = i(\infty) + (A_1 + A_2 t) e^{\alpha t}$$

Substitute 9 V for $i(\infty)$ and -2 for α .

$$i(t) = 9 + (A_1 + A_2 t) e^{-2t}$$

$$i(t) = 9 + (A_1 + A_2 t) e^{-2t} \dots\dots (1)$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= 9 + (A_1 + A_2(0)) e^{-2(0)} \\ &= 9 + (A_1 + 0) e^{(0)} \\ &= 9 + A_1(1) \\ &= 9 + A_1 \end{aligned}$$

Substitute 12 A for $i(0)$.

$$9 + A_1 = 12$$

$$A_1 = 12 - 9$$

$$A_1 = 3$$

Therefore, the value of A_1 is 3.

Step 12 of 13

Recall equation (1).

$$i(t) = 9 + (A_1 + A_2 t)e^{-2t}$$

Apply derivative with respect to t .

$$\frac{di(t)}{dt} = 0 + (A_1 + A_2 t)(-2)e^{-2t} + e^{-2t} A_2$$

Substitute 0 for t and 3 for A_1 .

$$\begin{aligned}\frac{di(0)}{dt} &= (3 + A_2(0))(-2)e^{-2(0)} + e^{-2(0)} A_2 \\ &= (3 + 0)(-2)e^{(0)} + e^{(0)} A_2 \\ &= (3)(-2)(1) + (1) A_2 \\ &= -6 + A_2\end{aligned}$$

Substitute 0 A/s for $\frac{di(0)}{dt}$.

$$A_2 = 6$$

Therefore, the value of the constant A_2 is 6.

Step 13 of 13

Recall equation (1).

$$i(t) = 9 + (A_1 + A_2 t)e^{-2t}$$

Substitute 6 for A_2 and 3 for A_1 .

$$\begin{aligned}i(t) &= 9 + (A_1 + A_2 t)e^{-2t} \\ &= 9 + (3 + 6t)e^{-2t} \text{ A}\end{aligned}$$

Therefore, the value of the current $i(t)$ is $9 + (3 + 6t)e^{-2t} \text{ A}$.

[Comments \(1\)](#)



Anonymous

Any idea why the textbook occasionally uses the unit step function next to the final answer? Is there any validity in it?